Ecole Supérieure d'Informatique et du Numérique
COLLEGE OF ENGINEERING & ARCHITECTURE

- TP 07: Cryptography
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- Niveau : 3ème année Ing

TP n° 07

Cryptography

1- Objectifs :

Cryptography is the science of encoding messages. It is common that one person (Alice) wants to communicate with a second person (Bob) so that a third party (Carl) cannot read the exchanged messages. There is a wide variety of ways to perform this encoding, and the most secure schemes rely on advanced mathematics.

The objectives of this lab are:

- To understand different classical encryption systems (substitution, Vigenère, transposition).
- To implement encryption and decryption using Google Colab / Jupyter Notebook (Python code cells).
- To perform basic **cryptanalysis**: recovering the plaintext or the key without prior knowledge, using tools like frequency analysis.

In this lab, we will focus on:

- Monoalphabetic substitution cipher (including Caesar cipher).
- Polyalphabetic substitution cipher (Vigenère cipher).
- Frequency analysis attack on substitution ciphers (cryptanalysis).

2- Security Properties

For each scenario, indicate which security property is violated:

Confidentiality, Integrity, Availability, Non-repudiation, Authenticity.

1. An attacker compromises a pharmaceutical factory and changes the chemical formula of the produced drugs.
2. An attacker steals the chemical formula of a drug.
3. An attacker breaks the authentication key to access the production systems.
4. An attacker stops the drug production system.
5. Propose a scenario to increase the security of your chemical formulas (countermeasure: think of encryption, access control, backups, digital signatures, etc.).

You can solve all exercises by:

- **First doing them by hand (on paper).**
- **Then implementing them in Python in Colab / Jupyter: one code cell per question or per cipher.**

Exercice 1 : Substitution Ciphers and Frequency Attack

1) Encrypt the plaintext securite using the Caesar cipher with key = 3 (shift each letter by 3 positions forward in the alphabet).
2) Encrypt the plaintext securite using the following monoalphabetic substitution key:

plaintext	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
ciphertext	N	L	E	H	X	Y	J	T	G	K	Z	D	A	R	W	M	C	Q	S	B	O	F	V	I	P	U

3) Give the names of two attacks that can break a monoalphabetic substitution cipher (examples: exhaustive key search, frequency analysis, known plaintext attack, etc.).
4) Decrypt the following monoalphabetic ciphertext, knowing that the key phrase is: « Il était une fois » (use it to construct the substitution key and then decrypt):

   HAWAEZXXIGHAD

5) Cryptanalysis – frequency attack:
   Decrypt the following sentence using **frequency analysis** (attack without knowing the key in advance)

   LHLZ HFQ BC HFFPZ WH YOUPFH MUPZH

   In Colab / Jupyter, implement:

   - A function to count the frequency of each letter in the ciphertext.
   - A display of sorted frequencies (histogram or simple list).
   - Use the typical English letter frequency (E, T, A, O, I, N, …) to guess substitutions and iteratively recover the plaintext.

Exercice2 : Vigenère Cipher (Decryption)

Decrypt the following message using the Vigenère cipher:

- Ciphertext: **vvqimsykttzbx**
- Key: **test**

Do this:

- Manually: write the Vigenère table or use modular arithmetic on letter indices.
- In Colab: write Python functions vigenere_encrypt (plaintext, key) and vigenere_decrypt (ciphertext, key) and verify that decryption recovers the original message.

Exercise 3 – Vigenère Cryptanalysis (Key Recovery)

You are given the plaintext and its Vigenère ciphertext:

- Plaintext: **MASTER TELECOM**
- Ciphertext: **AKGDSBHOZOQYA**

Tasks:

- Assume a Vigenère cipher.
- Compute, for each position, the shift between plaintext letter and ciphertext letter to deduce the key letters (remember to ignore spaces).
- Determine the **key** used.
- Verify your result in Colab by encrypting MASTER TELECOM with your recovered key and checking that you obtain AKGDSBHOZOQYA.

Exercise 4 – Transposition Cipher

1. Encrypt the message: **« seul celui qui voyage verra le long chemin qui mène à son pays »**

   using a columnar transposition cipher with the key: **désert** (remove accents for implementation: desert).

   Steps:

   - Remove spaces (and optionally accents) from the plaintext.
   - Write the text in rows under the key letters.
   - Sort the columns according to the alphabetical order of the key letters and read off column by column to get the ciphertext.

2. Decrypt the ciphertext:

   AGMBHNDJEKCIFL

   using the same key (desert).

   In Colab / Jupyter, implement:

   - A function transpose_encrypt (plaintext, key) for columnar transposition.
   - A function transpose_decrypt (ciphertext, key) to recover the plaintext.